



# Lecture Outline

- Classification: Why not Linear Regression?
- Binary Response & Logistic Regression
- Estimating the Simple Logistic Model
- Classification using the Logistic Model
- Multiple Logistic Regression

# Using Logistic Regression for Classification

How can we use a logistic regression model to perform classification?

That is, how can we predict when  $Y = 1$  vs. when  $Y = 0$ ?

We mentioned before, we can classify all observations for which  $\hat{P}(Y = 1) \geq 0.5$  to be in the group associated with  $Y = 1$  and then classify all observations for which  $\hat{P}(Y = 0) < 0.5$  to be in the group associated with  $Y = 0$ .

Using such an approach is called the standard **Bayes classifier**.

The Bayes classifier takes the approach that assigns each observation to the most likely class, given its predictor values.

# Using Logistic Regression for Classification

When will this Bayes classifier be a good one? When will it be a poor one?

The Bayes classifier is the one that minimizes the overall classification error rate. That is, it minimizes:

$$\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

Is this a good Loss function to minimize? Why or why not?

The Bayes classifier may be a poor indicator within a group. Think about the Heart Data scatter plot...



# Using Logistic Regression for Classification

This has potential to be a good classifier if the predicted probabilities are on both sides of 0 and 1.

How do we extend this classifier if  $Y$  has more than two categories?



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# Multiple Logistic Regression

It is simple to illustrate examples in logistic regression when there is just one predictors variable.

But the approach 'easily' generalizes to the situation where there are multiple predictors.

A lot of the same details as linear regression apply to logistic regression. Interactions can be considered. Multicollinearity is a concern. So is overfitting. Etc...

So how do we correct for such problems?

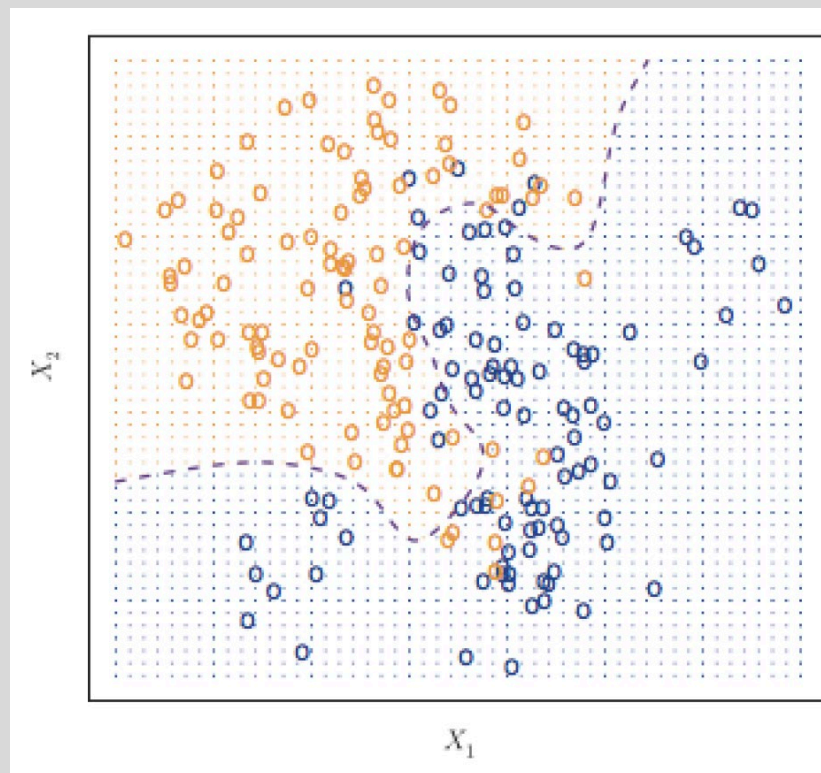
Regularization and checking though train, test, and cross-validation!

We will get into the details of this, along with other extensions of logistic regression, in the next lecture.



# Classifier with Two Predictors

How can we estimate a classifier, based on logistic regression, for the following plot?



# Multiple Logistic Regression

Earlier we saw the general form of *simple* logistic regression, meaning when there is just one predictor used in the model. What was the model statement (in terms of linear predictors)?

$$\log \left( \frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \beta_1 X$$

Multiple logistic regression is a generalization to multiple predictors. More specifically we can define a multiple logistic regression model to predict  $P(Y = 1)$  as such:

$$\log \left( \frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$



# Fitting Multiple Logistic Regression

The estimation procedure is identical to that as before for simple logistic regression:

- a likelihood approach is taken, and the function is maximized across all parameters  $\beta_0, \beta_1, \dots, \beta_p$  using an iterative method like Newton-Raphson.

The actual fitting of a Multiple Logistic Regression is easy using software (of course there's a python package for that) as the iterative maximization of the likelihood has already been hard coded.

In the `sklearn.linear_model` package, you just have to create your multidimensional design matrix  $X$  to be used as predictors in the `LogisticRegression` function.



# Interpretation of Multiple Logistic Regression

Interpreting the coefficients in a multiple logistic regression is similar to that of linear regression.

**Key:** since there are other predictors in the model, the coefficient  $\hat{\beta}_j$  is the association between the  $j^{th}$  predictor and the response (on log odds scale). But do we have to say? Controlling for the other predictors in the model.

We are trying to attribute the partial effects of each model controlling for the others (aka, controlling for possible *confounders*).

# Interpretation of Multiple Logistic Regression

Let's get back to the Heart Data. We are attempting to predict whether someone has HD based on MaxHR and whether the person is female or male. The simultaneous effect of these two predictors can be brought into one model.

Recall from earlier we had the following estimated models:

$$\log \left( \frac{\widehat{P(Y = 1)}}{1 - \widehat{P(Y = 1)}} \right) = 6.30 - 0.043 \cdot X_{MaxHR}$$
$$\log \left( \frac{\widehat{P(Y = 1)}}{1 - \widehat{P(Y = 1)}} \right) = -1.06 + 1.27 \cdot X_{gender}$$



# Interpretation of Multiple Logistic Regression

The results for the multiple logistic regression model are:

```
data_x = df_heart[['MaxHR', 'Sex']]
data_y = df_heart['AHD']

logreg = LogisticRegression(C=100000, fit_intercept=True)
logreg.fit(data_x, data_y);

print('Estimated beta1: \n', logreg.coef_)
print('Estimated beta0: \n', logreg.intercept_)
```

```
Estimated beta1:
[[-0.04496354  1.40079047]]
Estimated beta0:
[ 5.58662464]
```



# Interactions in Multiple Logistic Regression

Just like in linear regression, interaction terms can be considered in logistic regression. An **interaction term** is incorporated into the model the same way, and the interpretation is very similar (on the log-odds scale of the response of course).

Write down the model for the Heart data for the 2 predictors plus the interactions term.



# Interactions in Multiple Logistic Regression

The results for the multiple logistic regression model are:

```
df_heart['Interaction'] = df_heart.MaxHR * df_heart.Sex

data_x = df_heart[['MaxHR', 'Sex', 'Interaction']]
data_y = df_heart['AHD']

logreg = LogisticRegression(C=100000, fit_intercept=True)
logreg.fit(data_x, data_y);

print('Estimated beta1, beta2, beta3: \n', logreg.coef_)
print('Estimated beta0: \n', logreg.intercept_)

Estimated beta1, beta2, beta3:
[[-0.02645985  5.38749287 -0.02689767]]
Estimated beta0:
[ 2.88218441]
```



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# Questions?